Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/30/2021
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Johns Hopkins University		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	02/07/2019	**End Date:**	02/07/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/7/2033 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/18/21) NOTE: End date changed to 6/6/2021 per NSSC information (Ed., 11/9/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	80NSSC19K0487		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose a simple and rapid measure of otolith asymmetry -- a basic low-level neurovestibular function that is altered in spaceflight. Our assessment apparatus and procedure was developed with previous Human Research Program (HRP) support, and has undergone validation testing and is currently in use in an unrelated Department of Defense (DoD) study. It provides a measure of binocular alignment (vertical and torsional), which is a reflection of underlying otolith asymmetry. While this measure might be important in itself (since it reflects a form of central neural compensation that might be challenged in spaceflight), it could also provide a simple alternative to locomotion and posture testing if it is found to correlate with those measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology and procedures for this project are being used in DoD projects for the assessment of vestibular function in military personnel. The technology has potential clinical applications.
Task Progress:	Progress since November 2020 This project was selected for flight (SFF), in modified form as noted below, in October 2020. Ocular-alignment assessment has been incorporated into the Vestibular Health Complement of Integrated Protocols for Human Research (CIPHER) project. Discussions continue with Scott Wood, who represents that project. The combined project uses a commercial head-mounted video system to measure eye movements. The original tablet-computer version of the test will be used for the first preflight test session as a backup; it can be used in place of the more complex head-mounted video system as a contingency. Institutional Review Board (IRB) approval, in the form of a Reliance Agreement, was finalized between NASA Johnson Space Center (JSC) and Johns Hopkins University (JHU), for the tablet-computer version of the test. The JHU team will be added to the Vestibular Health IRB protocol, and the JSC team will be added to the JHU protocol. NASA IRB approval was obtained, including multinational consenting. In 2021, we supported online integrated consent briefings, as well as a combined Test Readiness Review (TRR), and finalized the Science Requirements Document. A presentation was made at the online HRP Investigators Workshop in February 2021. While awaiting CIPHER flight opportunities, a portion of the protocol for this project was modified and extended for use as an educational demonstration at Space Center Houston (during which relevant baseline data will be obtained), and as a pre/post-flight test for commercial spaceflights (under separate Translational Research Institute for Space Health (TRISH) support). Recently (fall 2021), we started a set of experiments in the lab, to provide further baseline data by which to evaluate the upcoming flight data from this project. In these trials, subjects will be evaluated for ocular alignment (OA) while standing on a hard floor, standing on foam, and while seated. This will help us evaluate the impact of postural demands on ocular alignment; we conjecture that the heightened demand while standing will call for a more precise compensation for any latent otolith asymmetry, causing a reduction in misalignment in that condition compared to seated.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Shelhamer M, Schubert M. "Assessment of otolith function and asymmetry as a corollary to sensorimotor performance in missions of various durations and as an investigation of partial-g effects." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021